E-Commerce Demand Forecasting

Predict future product demand to optimise inventory and reduce stockouts & overstocking.

Project Number - SG5

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Problem Statement

What We're Solving

Predict future product demand to optimise inventory and reduce stockouts & overstocking.

Business Questions That Will be Answered

- How many units of each product should be stocked next week/month?
- Which products show strong seasonality?
- How do promotions & holidays impact demand?
- How accurate are ML vs deep learning models?

About Dataset

Kaggle – Store Item Demand Forecasting Challenge

Features

- date
- store_id
- item_id
- sales

Why this dataset?

- Time-series data
- Multiple stores
- Multiple items
- Seasonal patterns
- Trend patterns
- Cross-sectional structure (store × item combinations)

XGBoost & LSTM

Two Powerful Approaches

XGBoost (Extreme Gradient Boosting)

XGBoost (Extreme Gradient Boosting) is an advanced tree-based ensemble algorithm based on gradient boosting. It builds many decision trees sequentially, where each new tree corrects the errors made by the previous trees. Instead of training all trees independently, it minimises a loss function using gradient descent. It includes regularisation to prevent overfitting and supports parallel computation, making it fast and efficient. XGBoost is especially powerful for structured/tabular datasets with engineered features. It handles non-linear relationships and feature interactions very effectively.

LSTM (Long Short-Term Memory)

LSTM (Long Short-Term Memory) is a type of Recurrent Neural Network (RNN) designed for sequential data. It processes data step-by-step whilst maintaining a memory of previous inputs. Unlike normal RNNs, LSTM uses gates (input, forget, output gates) to control what information to remember or discard. This helps it learn long-term dependencies without vanishing gradient problems. It is widely used for time-series forecasting, speech recognition, and NLP. LSTM is ideal when past values strongly influence future predictions.

Why Choose These Models?

Why Choose XGBoost for This Project?

For the e-commerce demand forecasting dataset:

- Data is structured (store, item, date, sales).
- We can create lag features, rolling means, month, weekday features.
- XGBoost captures complex feature interactions.
- It trains faster and is easier to tune compared to deep learning.

Why Choose LSTM for This Project?

- Sales data is sequential (time-series).
- Demand depends on previous days' sales.
- LSTM captures temporal dependencies automatically.
- It learns trends and seasonality from sliding windows.
- Good for modelling long-term patterns in demand.

Progress Till Now

1. EDA Conclusion

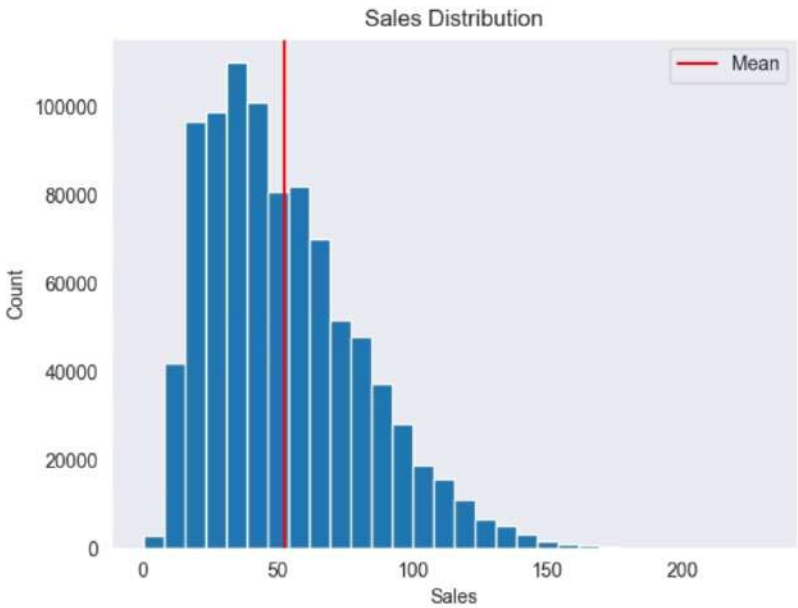
- The dataset is a clean time-series dataset with daily sales records from 2013 to 2017, containing 9,13,000 rows and 4 columns. Dataset includes the features date, store (10 unique stores), item (50 unique items), and sales, with no missing or duplicate values.

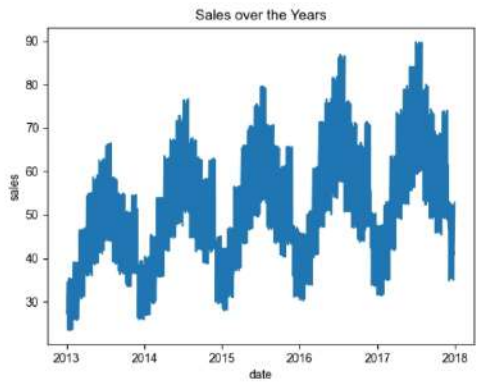
- Sales exhibit a clear upward trend over the years, indicating increasing demand over time.

  And Strong yearly seasonality is observed, with repeating demand patterns across years, confirming that sales are non-stationary.

- Most daily sales values fall within the range of 0–100 units, representing regular demand levels across stores and items.

- All stores contain outliers, indicating that these high-sales values represent genuine demand spikes rather than data errors; therefore, outliers should be retained, and store-specific modelling is likely to improve forecasting performance.





2. Feature Engineering Conclusion

Before there were 4 features, And now there are 19 features in total including—

Time features

Seasonality

Lag features

Memory

Rolling features

Trend & volatility

Aggregate features

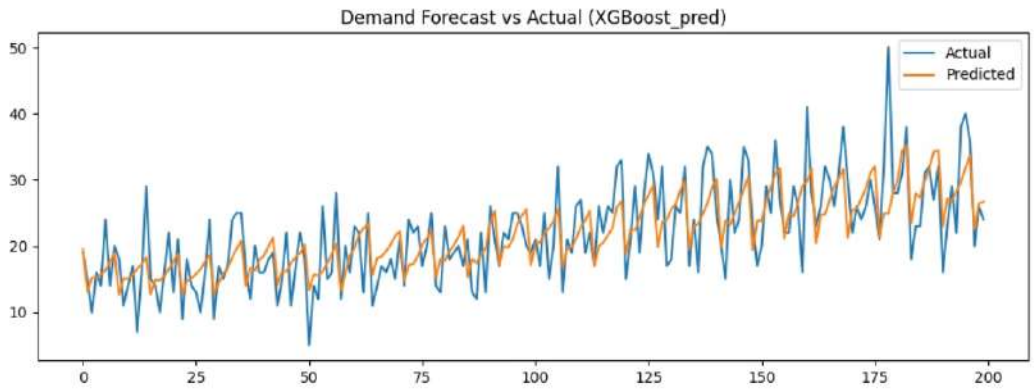
Indicate-Behaviour

Done with XGBoost Implementation

RMSE: 7.91

On average our forecast is off by ~8 units of sales (reasonable)

MAPE: 12.43%



What's Next

Future Plans

Implement Prophet

Alongside LSTM and XGBoost

Comparative Evaluation

Select the best-performing model

Develop GUI Application

For real-world demand forecasting usage

Future work includes implementing Prophet alongside LSTM and XGBoost, performing comparative evaluation to select the best-performing model, and finally developing a GUI-based application for real-world demand forecasting usage.